Group 1 | 03/28/2026

Sprint 2 Project

FILIPINO FAMILY INCOME AND EXPENDITURE

Creating a predictive model on poverty based on 2015 FIES Data

PRESENTED BY:

Marl Jay Ramos
Ezekiel Adriel Lagmay
Aiji Umemori
Maxine Nicole Bernales

Presentation Flow

Problem Statement

Many Filipino families **struggle to meet their daily needs** due to **limited income**.

...thus, **identifying which households are at risk of poverty** is essential for effective support and policy planning.

What we have

2015 Dataset of Filipino Family Income and Expenditure

What is the state of the Philippines in 2015?

In 2015, the average annual income of Filipino families was P268,000, while average expenditure reached P205,000.

Around 16.5% of families lived below the poverty line.

Source: Philippine Statistics Authority
(Family Income and Expenditure Survey, 2015)

What we have

2015 Dataset of Filipino Family Income and Expenditure

To **classify household poverty status** based on income and expenditure patterns

Our Goal

Why is this important?

- helps identify vulnerable households
- support better targeting of government programs
- reveals patterns in income and spending behavior

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Scope

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- Evaluation of Supervised Machine Learning models

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Limitations

Limited to 2015 data, so it does not reflect economic changes.

Why is this STILL important?

Even though the data is from 2015, it helps us understand the **baseline behavior** of Filipino households before major economic changes.

TARGET SDG

1 NO POVERTY

End Poverty in All its Forms Everywhere

#2: DATASET OVERVIEW AND PREPARATION

Dataset Overview

Source: Kaggle

Title:

Filipino Family Income and Expenditure

(Annual Household Income and Expenses in the Philippines)

Data Collection Method:

Computer-Assisted Personal Interview (via Tablet), 4 hours over two visits

Raw Dataset Overview

Before pre-processing

41,544

data points

Size

Raw Dataset Overview

Before pre-processing

41,544

data points

Size

60

columns

Feature Count

#3: Exploratory Data Analysis

Raw Dataset
Descriptive Stats

Shape: (41544, 60)

Null Values:

- Household Head Occupation - 7536
- Household Head Class of Worker - 7536
- Toilet Facilities - 1580

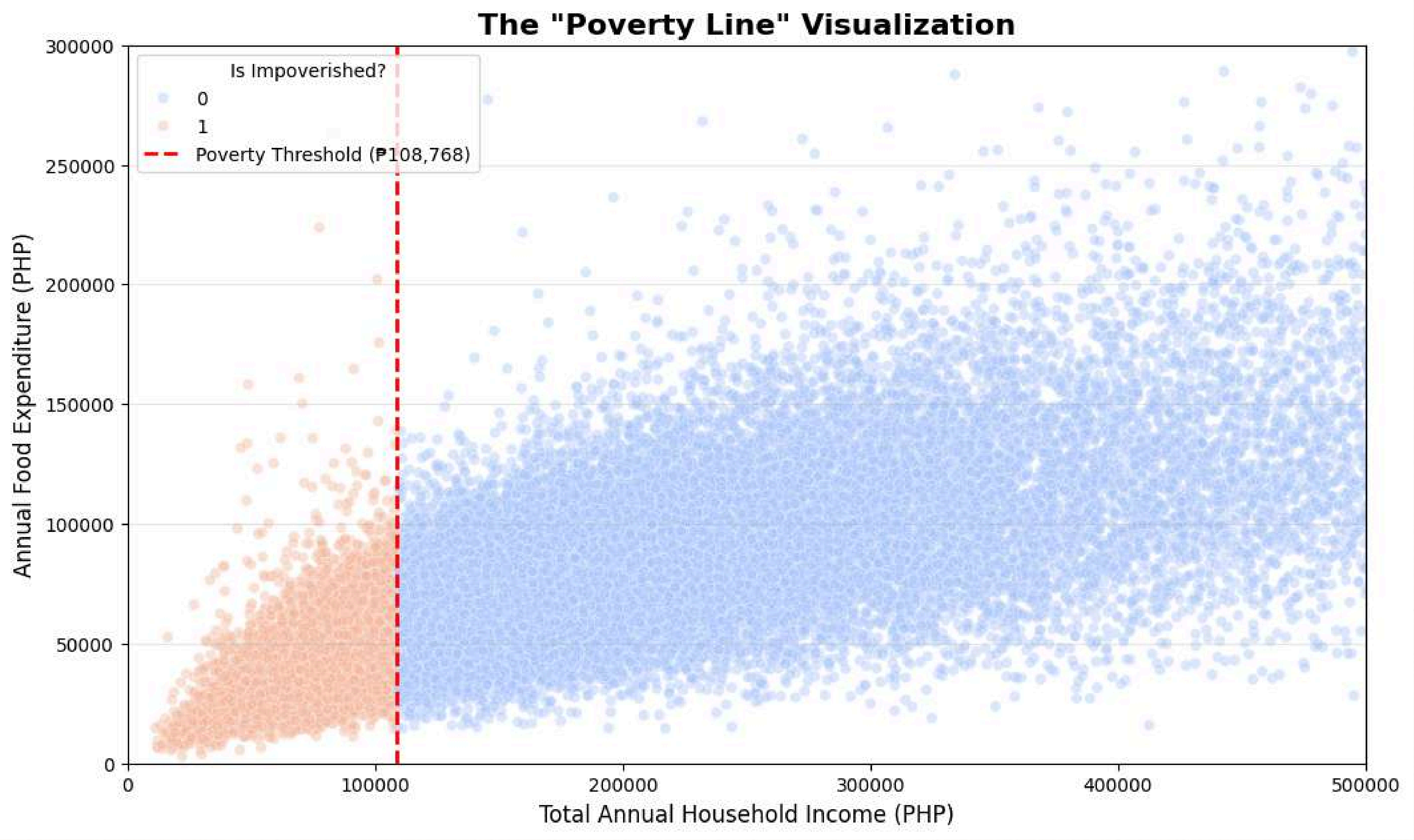
Duplicated Rows: 0

Data Preparation
Dealing with the mess

For Null Values: Change cells with null values

1. Household Head Occupation → “No occupation”
2. Household Head Class of Worker → “Unemployed”
3. Toilet Facilities → “Not reported”

A household is considered impoverished when their total annual household income is below P108,768.



Data Preparation
Feature Generation

is_impoverished

True

THI < P108, 768

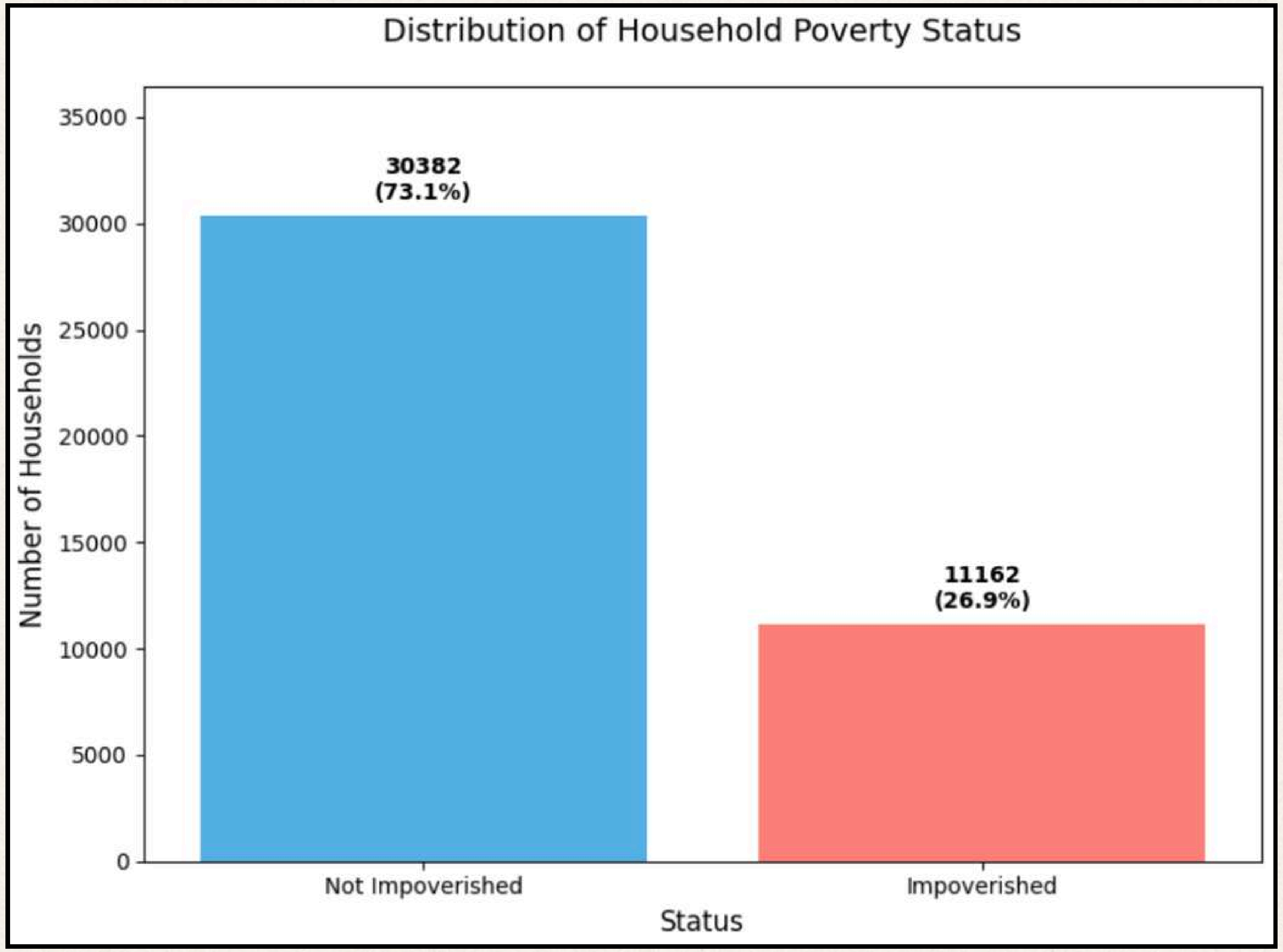
False

THI >= P108, 768

THI = Total Household Income

Household = generally represented by a family of 5
(Poverty Threshold: P9,064 income per month)

Dataset Overview
Target Variable Distribution

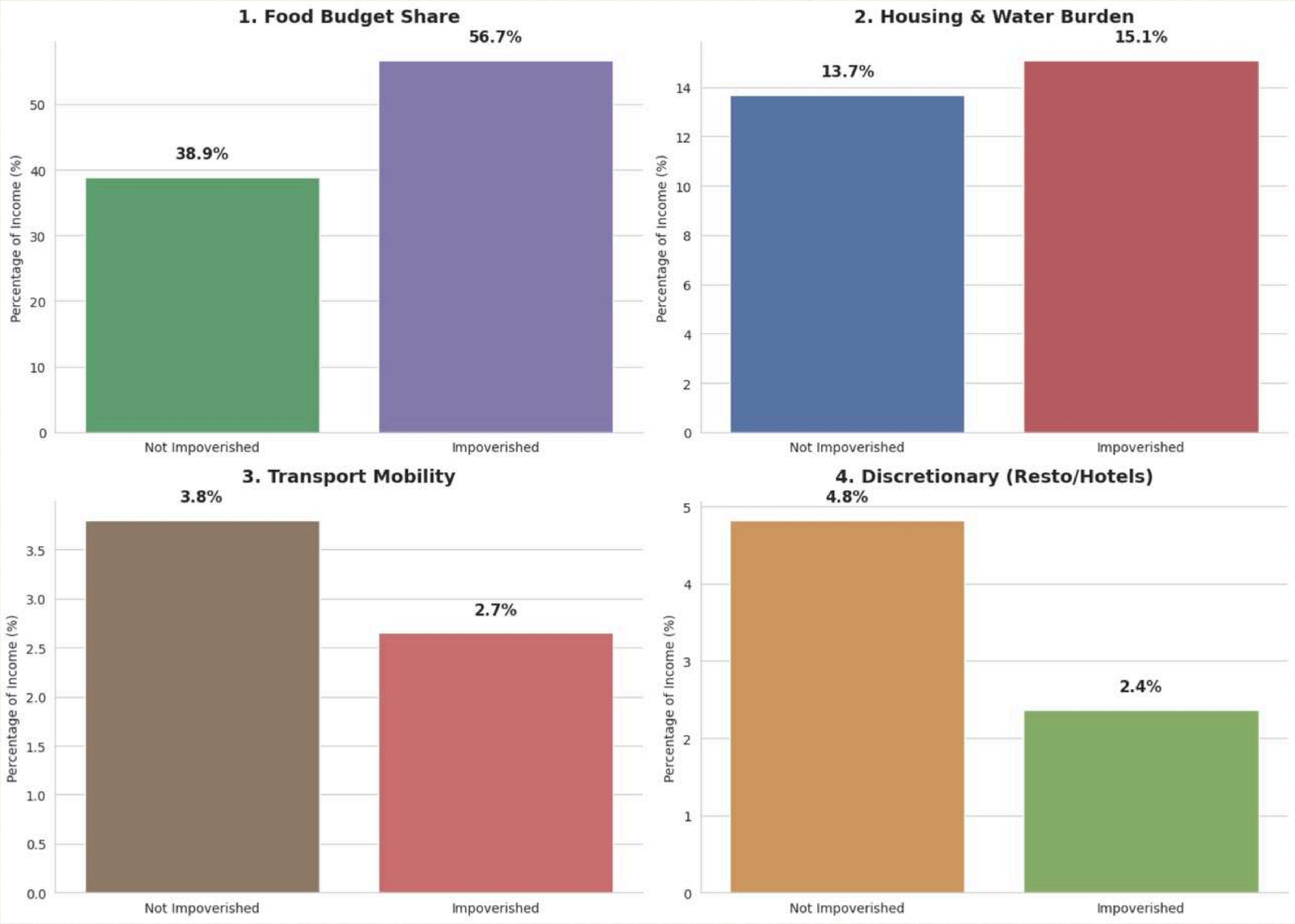


Minority class: Impoverished

< 30% of the total

Median Annual Income:
Not Impoverished: ₱216,537.50
Impoverished: ₱80,173.00

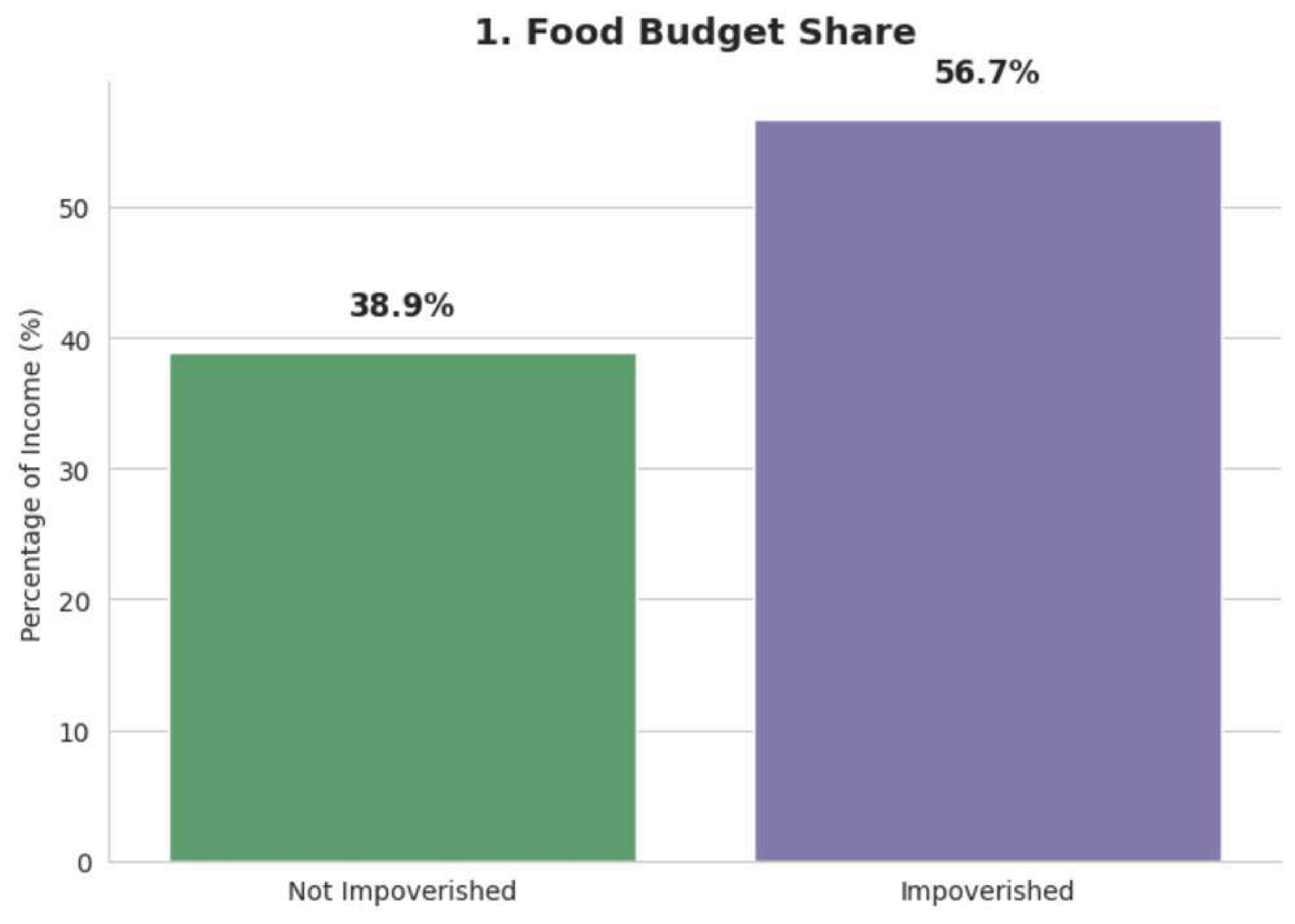
Topline EDAs:
Expense Snapshot relative to Income



Insight: Much emphasis is placed on necessities for impoverished households, to which the features represent a higher part on their income allocations compared to not impoverished households.

Topline EDAs:

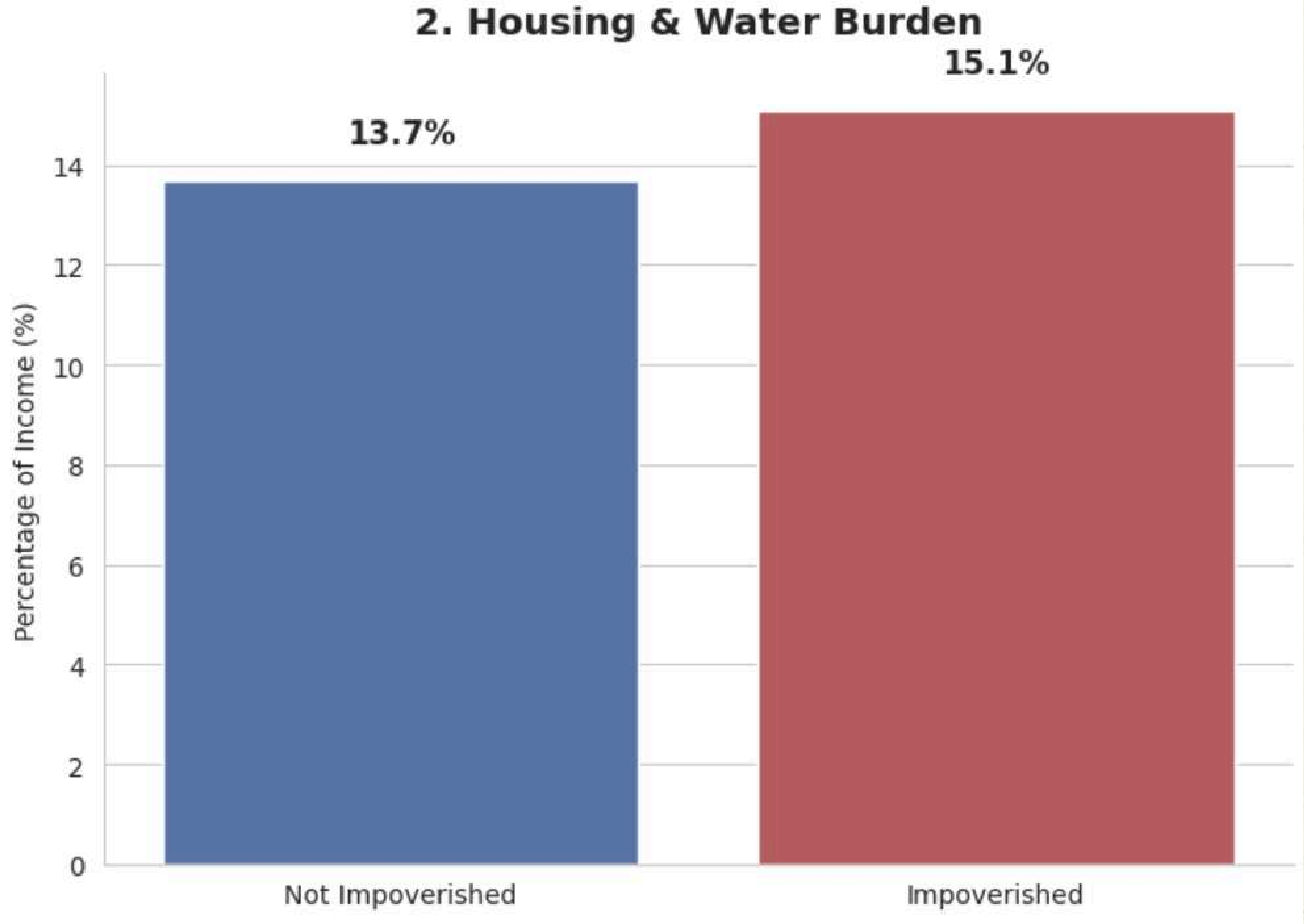
Expense Snapshot relative to Income



As a necessity, impoverished households allocate almost 20% much more of their incomes on food. This signals a bigger emphasis on survival for prioritization.

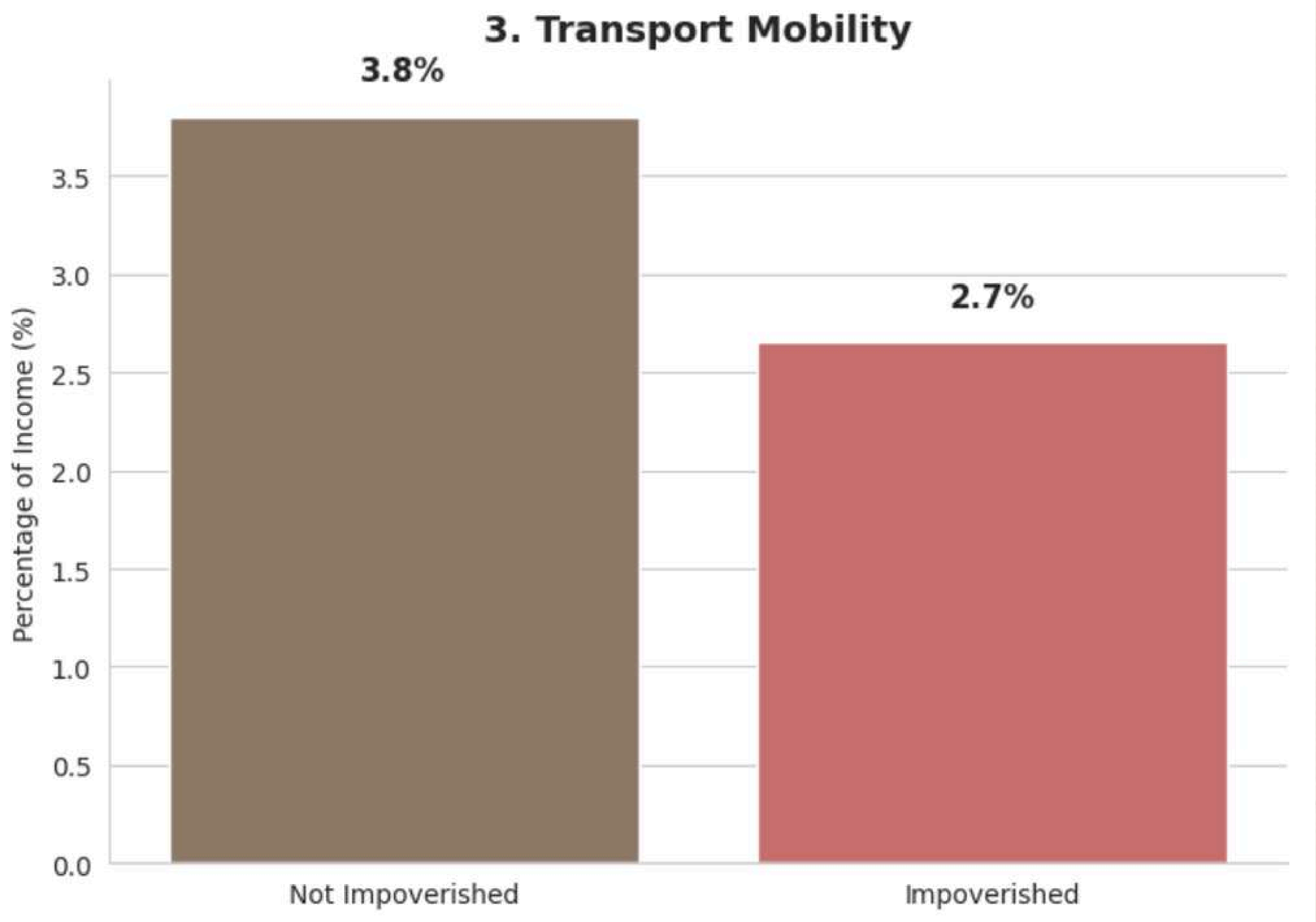
Topline EDAs:

Expense Snapshot relative to Income



There is a marginal difference in the way households allocate for housing and water, which are often seen as fixed expenses.

Topline EDAs:
Expense Snapshot relative to Income



Not impoverished groups allocate higher for transportation, which may be indicative of their chosen methods of transportation.

Fun fact: GrabCar started in 2015.

Unseen costs: comfort and time savings

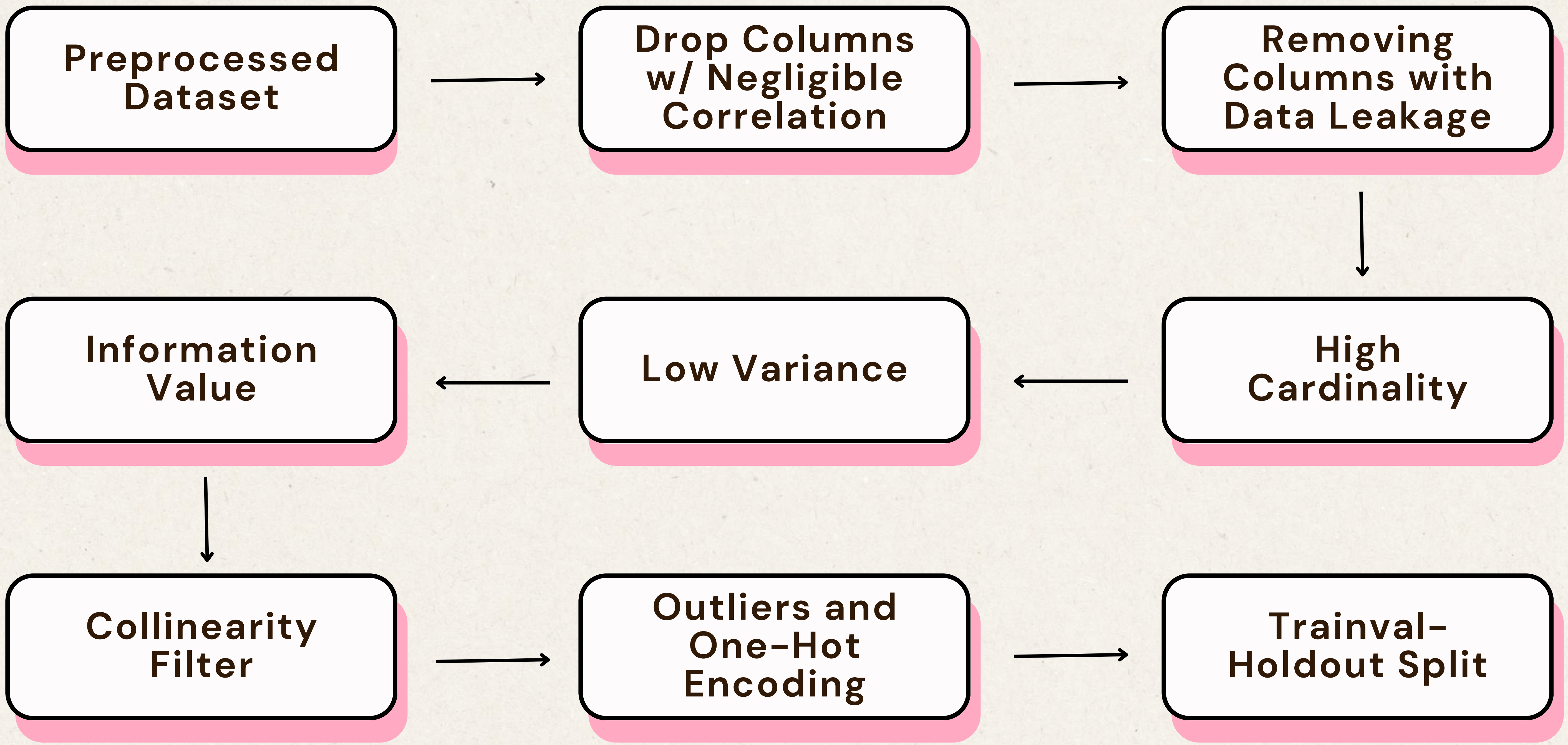
Topline EDAs:
Expense Snapshot relative to Income

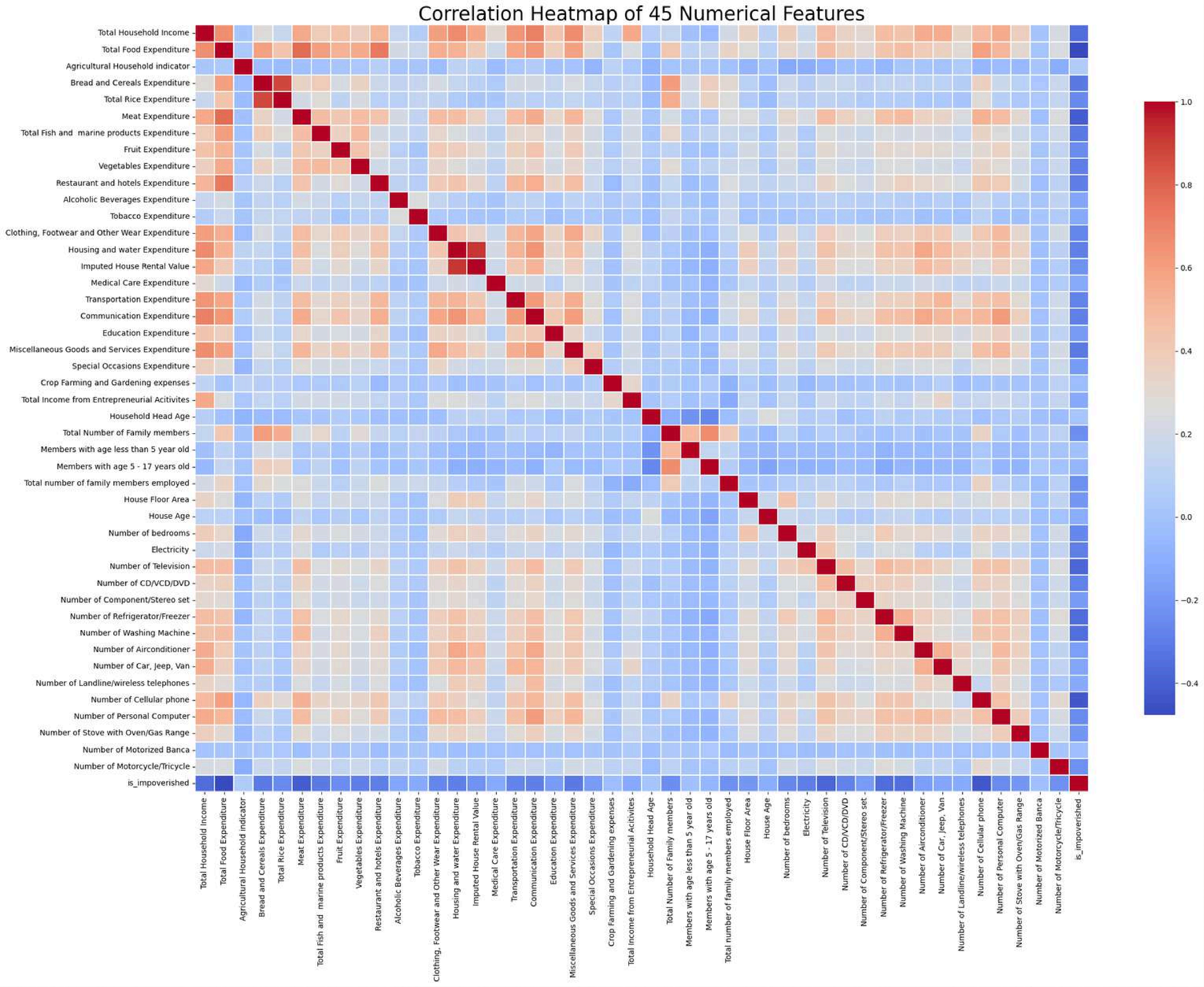
Not impoverished households allocate twice the percentage of their incomes on discretionary spending. As proven by previous graphs, there is less pressure on their budgets with regard to necessities.

See also: lifestyle inflation

#4: Feature Selection

Feature Selection
Process Flow





Low-Signal Filters: Using this correlation heatmap, **columns with negligible correlation will be dropped**.

$|r| < 0.2$

Feature Selection
Process Flow

Drop Columns w/ Negligible Correlation	61 → 47, 14 columns dropped	
Removing Columns with Data Leakage	47 → 46, 1 column dropped	
High Cardinality	46 → 43, 3 columns dropped	
Low Variance	43, 0 columns dropped	
Information Value	43 → 39, 4 columns dropped	
Collinearity Filter	39 → 34, 5 columns dropped	
Outliers and One-Hot Encoding	41,544 → 35,518 6,026 rows dropped	34 → 80, 4 columns dropped
Trainval-Holdout Split	**Trainval:** 27,209 rows, 80 columns	**Holdout:** 8309 rows, 80 columns

Final Dataset Overview

After pre-processing

35,518

data points
(80% Trainval, 20% Holdout)

Size

is_impoverished

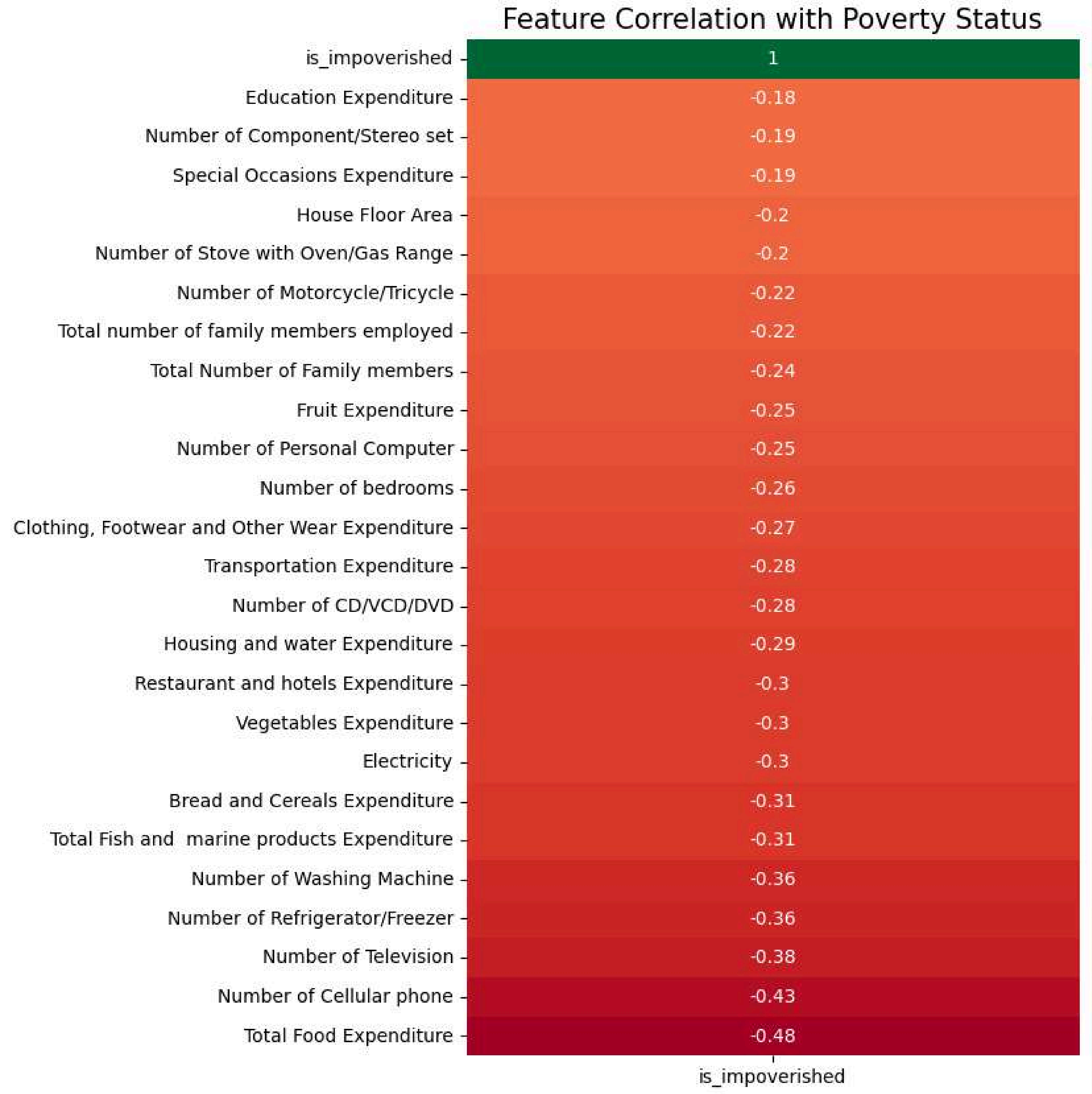
(1: yes, 0: no)

Target Variable

79 + 1

columns

Feature Count



The **top features** that correlated to household's poverty status are:

Total Food Expenditure

Devices and Appliances

Phone

Television

Fridge

Washing Machine

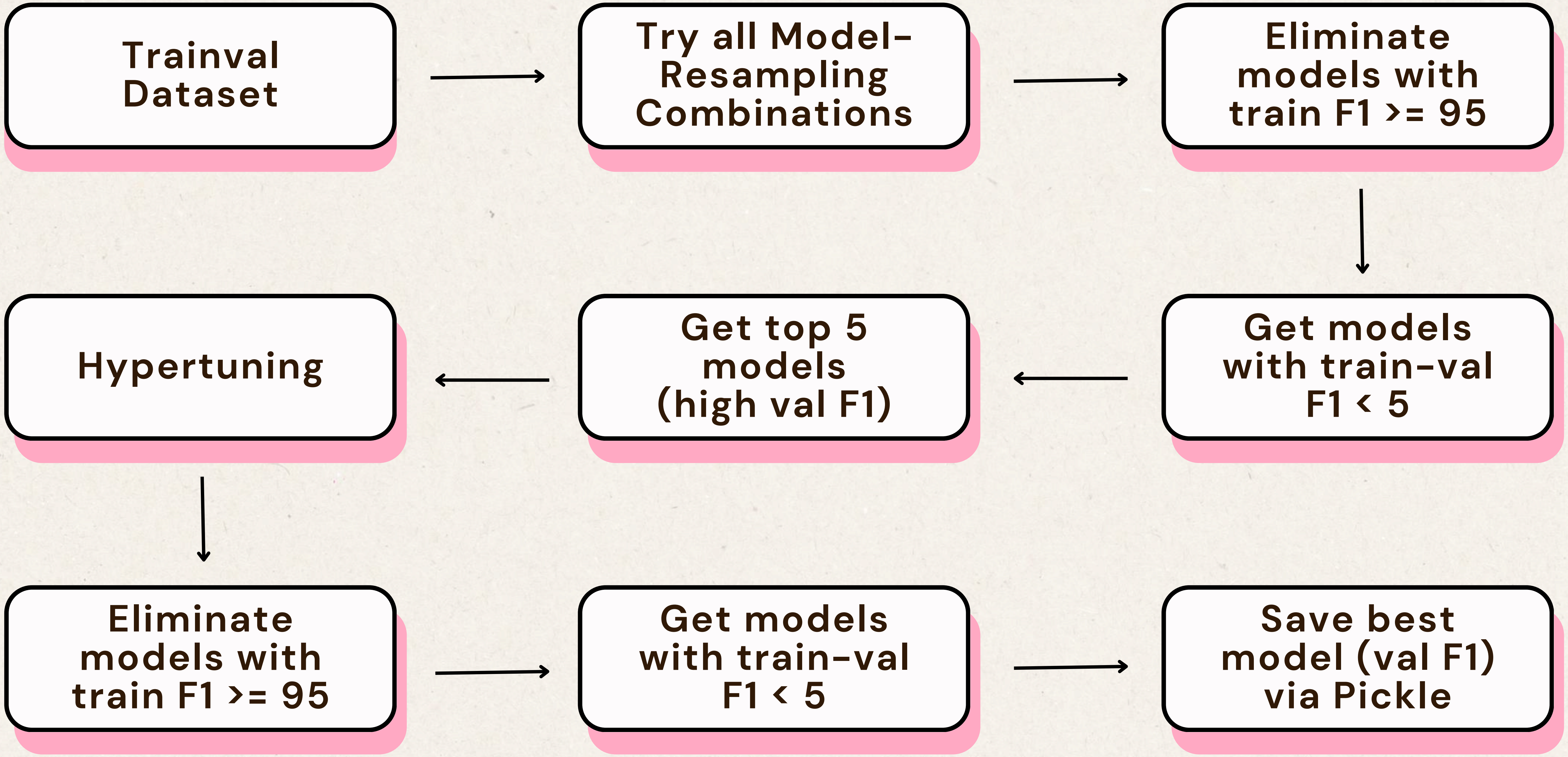
Grocery Expenditure

Fish and Marine

Bread and Cereals

#5: Model Choice and Performance Metric

Model Choice
Process Flow



Metrics 101

Precision

- The number of positive class predictions which actually belongs to the positive class.
- True Positives / (True Positives + False Positives)

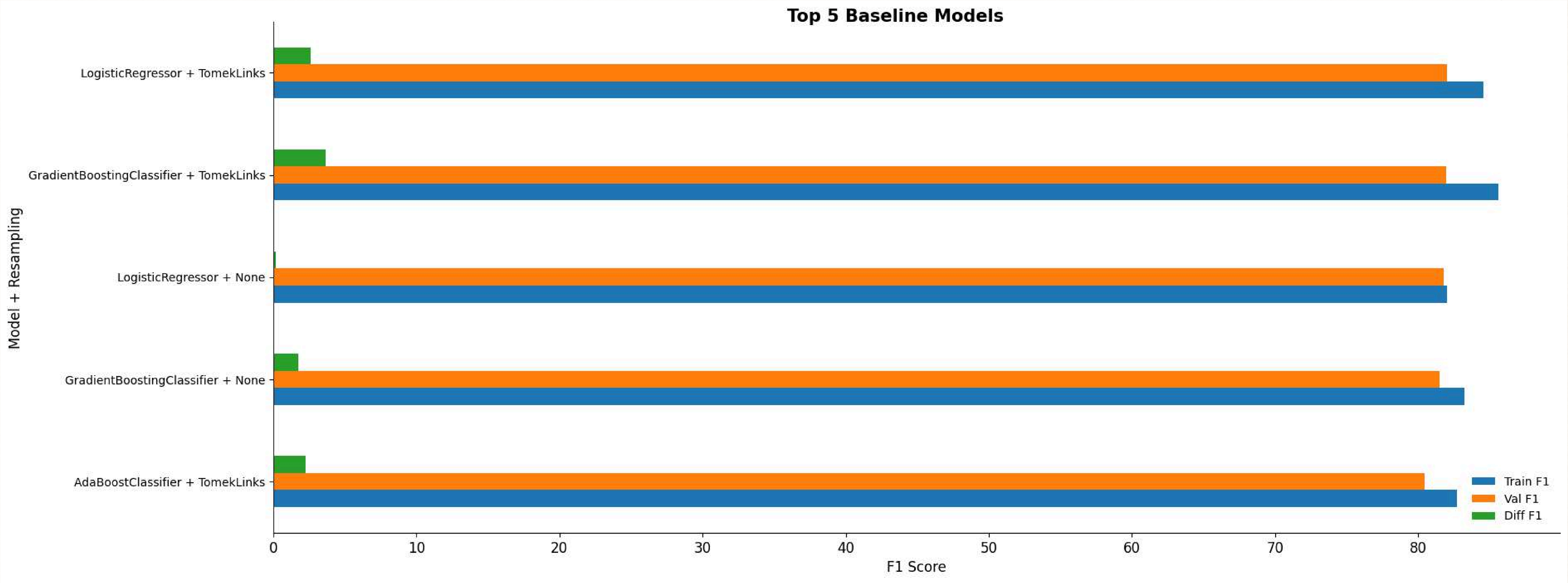
Recall

- Ability to predict positive samples.
- True Positives / (True Positives + False Negatives)

F1-Score

- Harmonic mean of Precision and Recall
- 2 * (Recall * Precision) / (Recall + Precision)

The Top 5 Baselines



Hyperparameters 101

Logistic Regression

- C = [0.1, 1.0, 10.0] (regularization strength → lower values = less overfitting)

Gradient Boosting Classifier

- Learning Rate = [0.1, 1.0, 10.0]
- N Estimators = [3, 4, 5]
- Max Depth = [3, 4, 5]

Ada Boost Classifier

- Learning Rate = [0.1, 1.0, 10.0]
- N Estimators = [3, 4, 5]

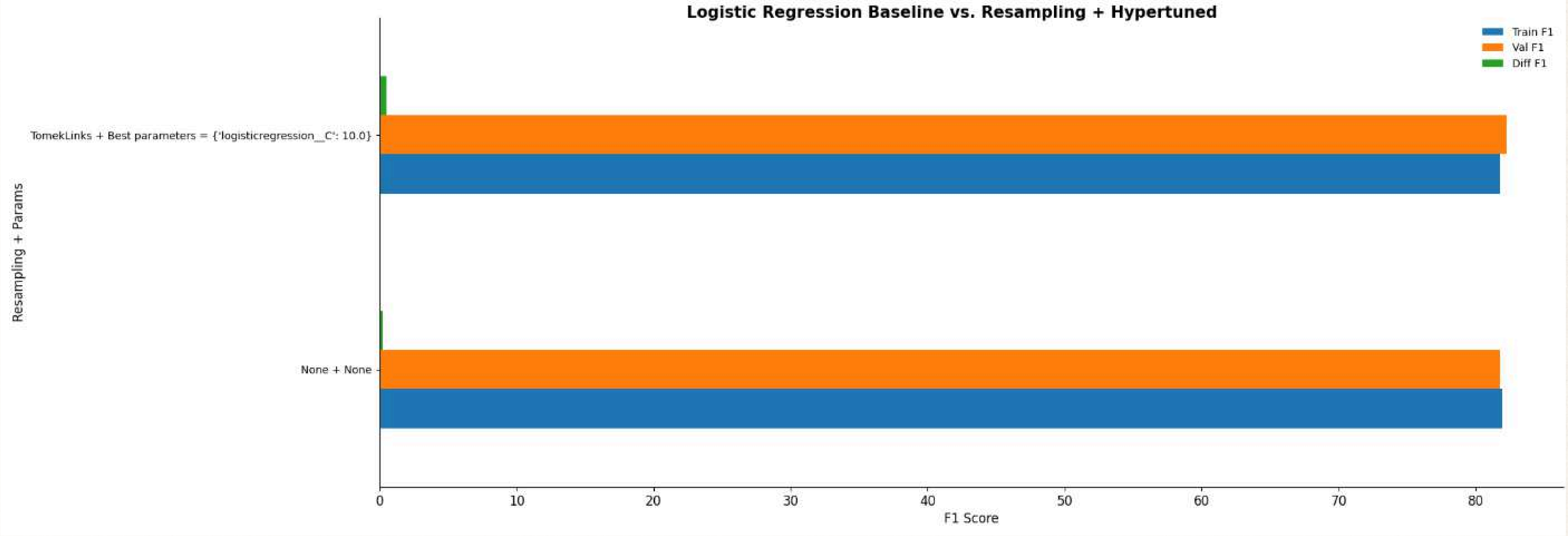
The Chosen One

Logistic Regression + TomekLinks

- Logistic Regression = Fits an S-curve to the observations.
- C = 10.0
- TomekLinks = Undersampling which removes data points from the majority class which are very close to the data points from the minority class.

Source: https://help.sap.com/docs/SAP_HANA_PLATFORM/2cfbc5cf2bc14f028cfbe2a2bba60a50/aad0816b1f434774957717206d8d350c.html?version=2.0.07

The Chosen One

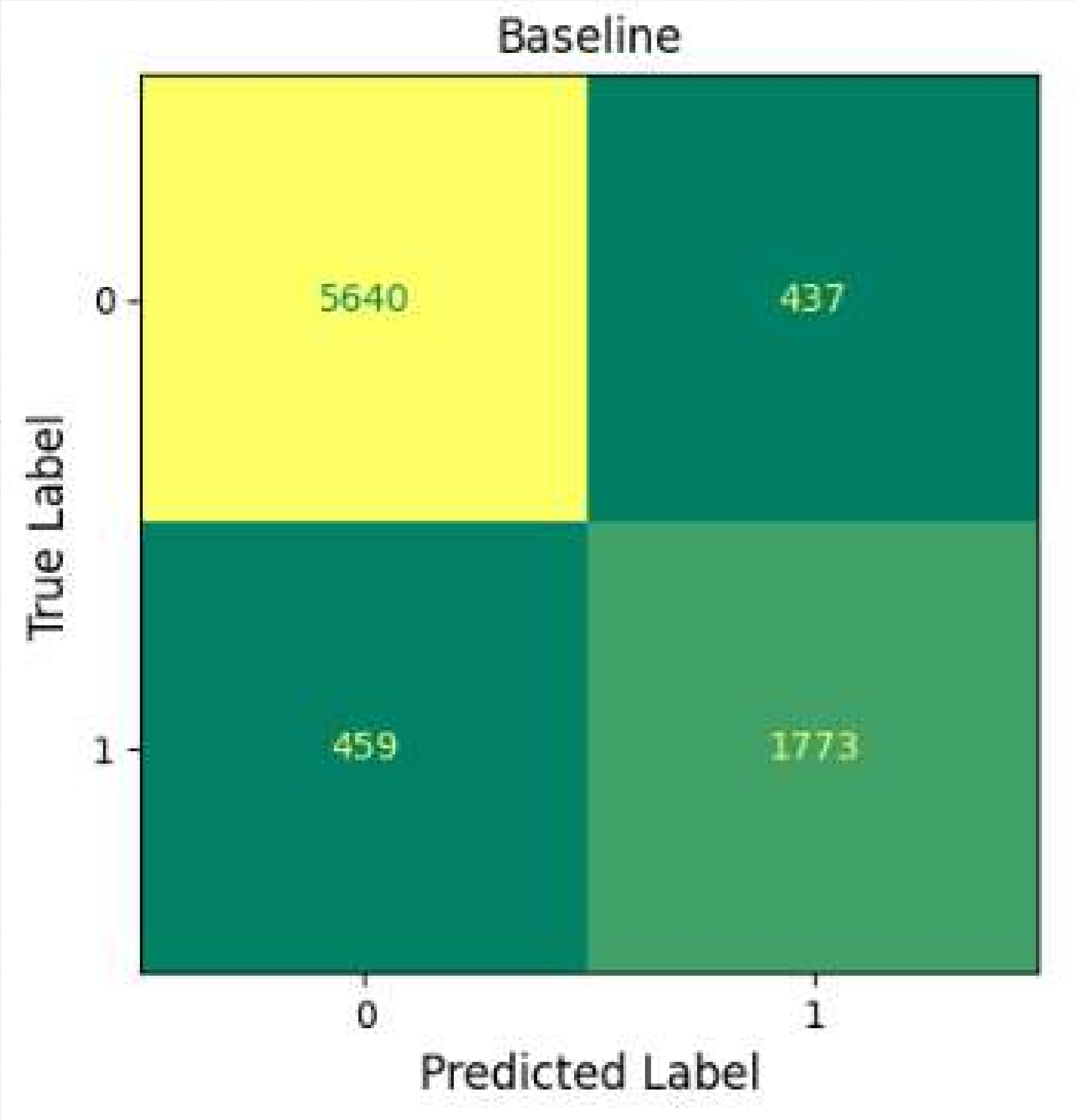


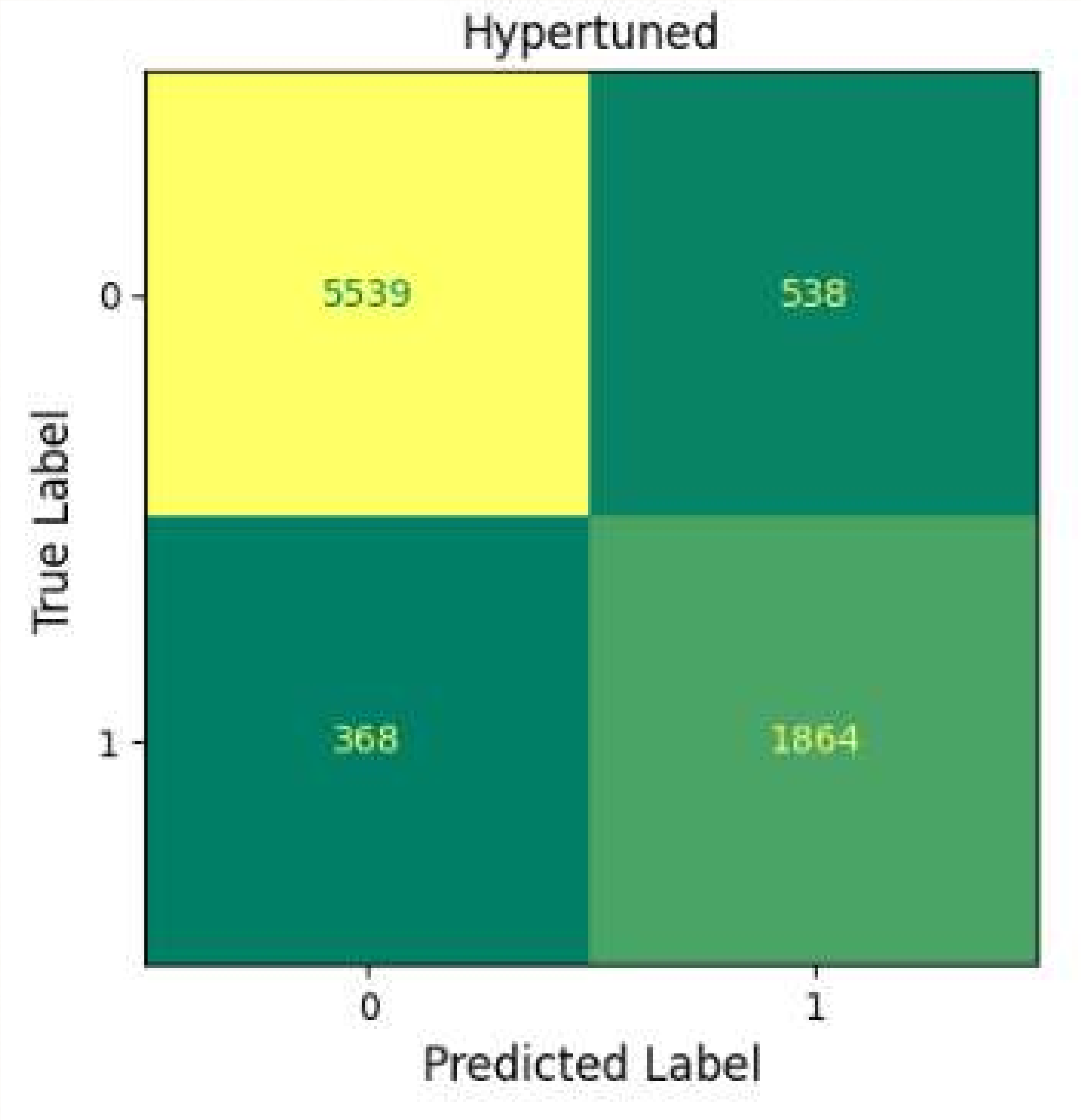
#6: Model Evaluation and Interpretation

Final Trained Models

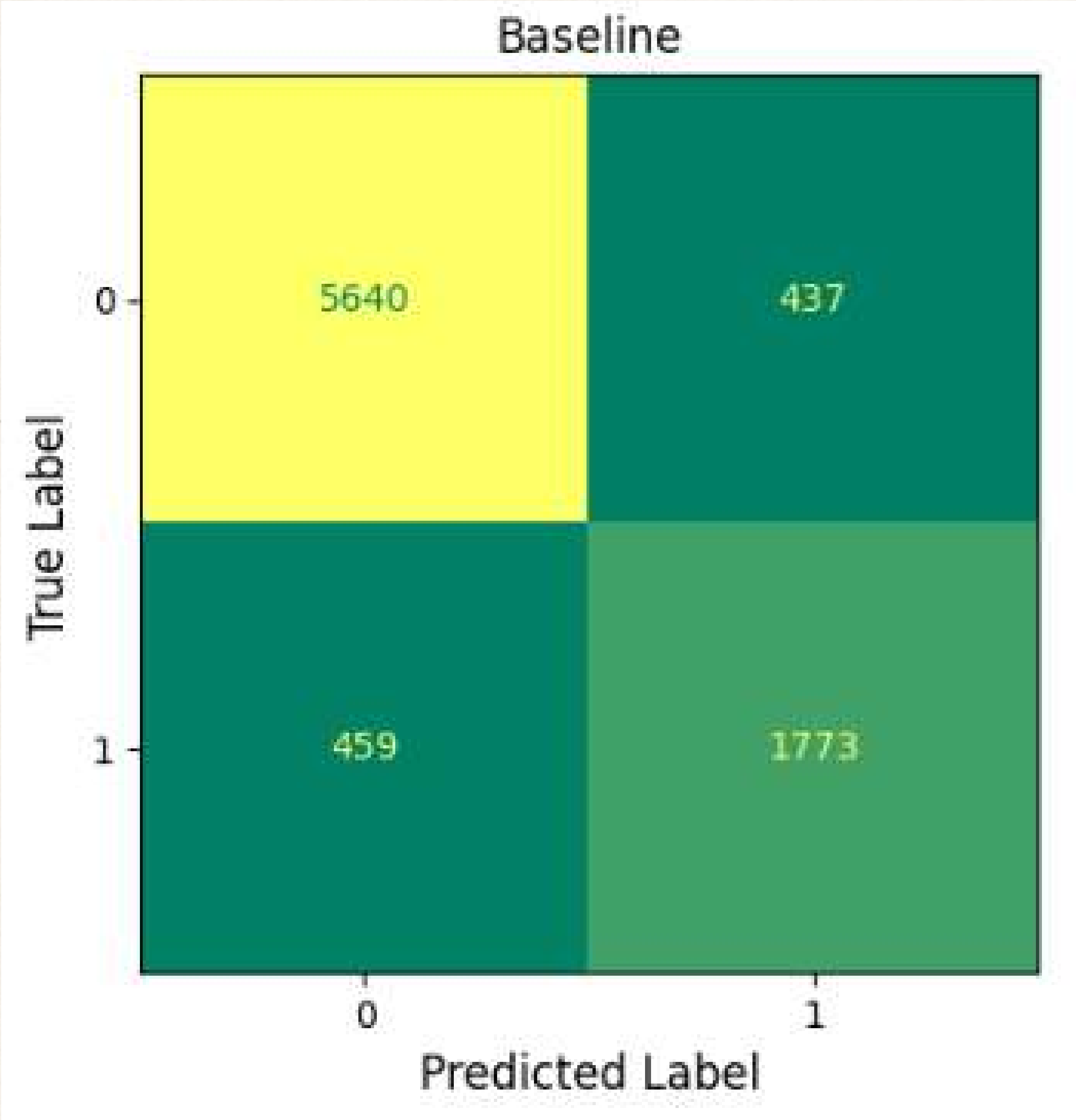
1. Baseline Model
2. Hyperparameter Tuned Model

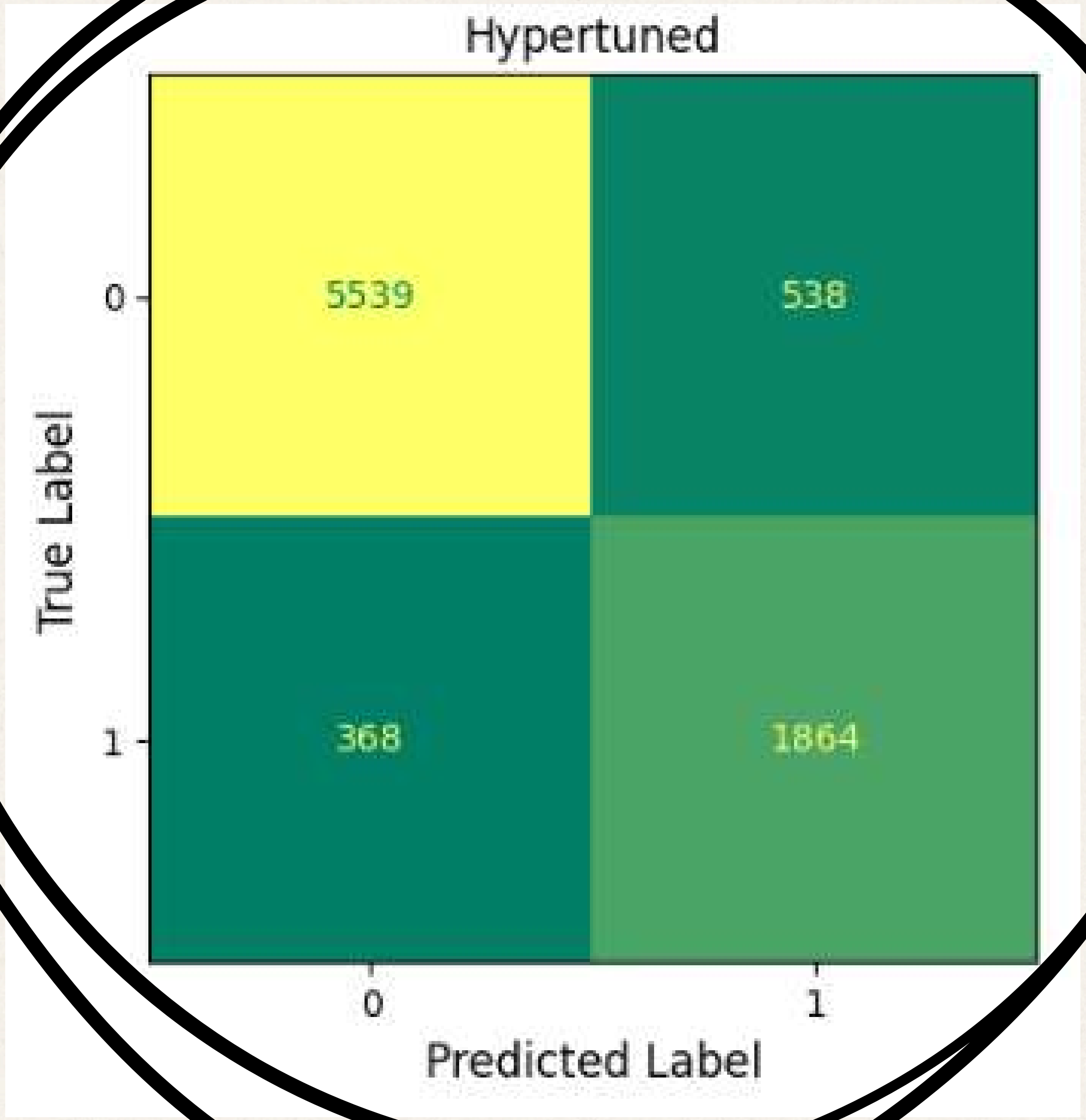
Holdout Confusion Matrix





Holdout Confusion Matrix





Holdout F1 Scores

75.88%

Proportion Chance Criterion * 1.25

79.83%

Baseline

80.45%

Hypertuned

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75.88%

Proportion
Chance Criterion
* 1.25

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Hypertuned

Model Explainability

SHAP (Shapely Additive Explanation)

- Higher Shapley Values for a feature → More Relevant
- Kernel SHAP - for Logistic Regression

"Which features matter most overall?"

LIME (Local Interpretable Model-agnostic Explanations)

- Creates a surrogate model and determines which features are the most affected.

"Why did the model predict THIS for this particular household?"

Baseline Model

SHAP Analysis

- **Expenditure patterns** are more predictive than income source or demographic.

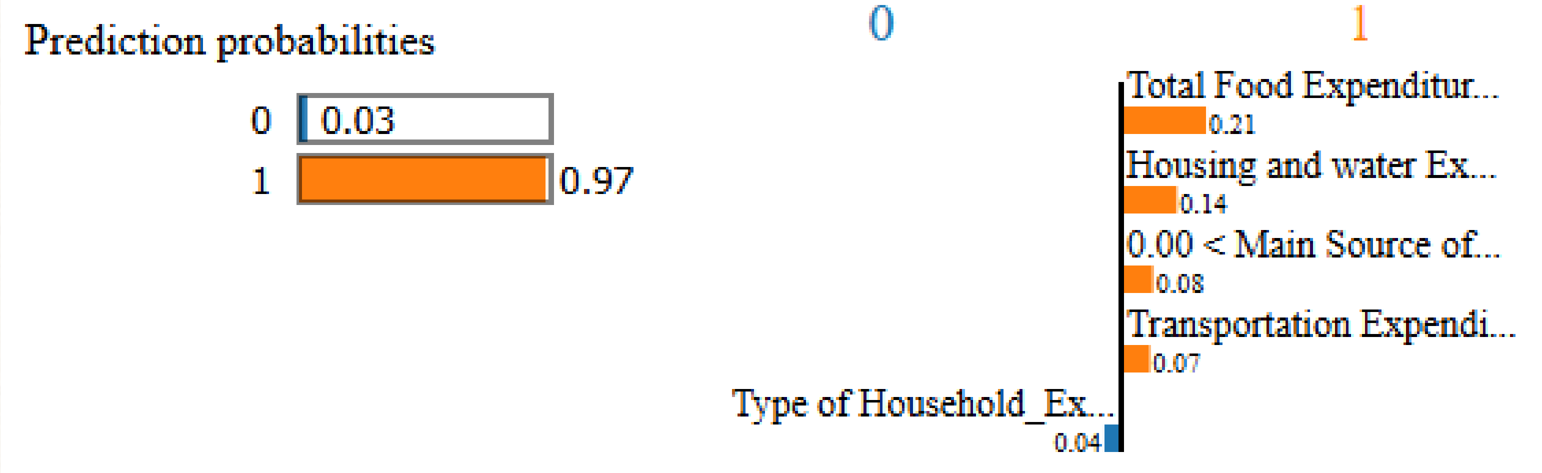
  Total Food and **Housing & Water** Expenditure dominate as the two strongest predictors

- **Demographic features** have moderate-to-low importance.

Baseline model leans heavily on expenditure patterns over household characteristics

Baseline Model

LIME Analysis



Feature	Value
Total Food Expenditure	27688.00
Housing and water Expenditure	11772.00
Main Source of Income_Other sources of Income	1.00
Transportation Expenditure	1278.00
Type of Household_Extended Family	0.00

In this household:

The model is **97%** confident this household belongs to Class 1 because its **food and housing spending patterns** strongly match that class profile, AS WELL AS the **transportation spending** and **other sources of income**

Hyperparameter Tuned Model

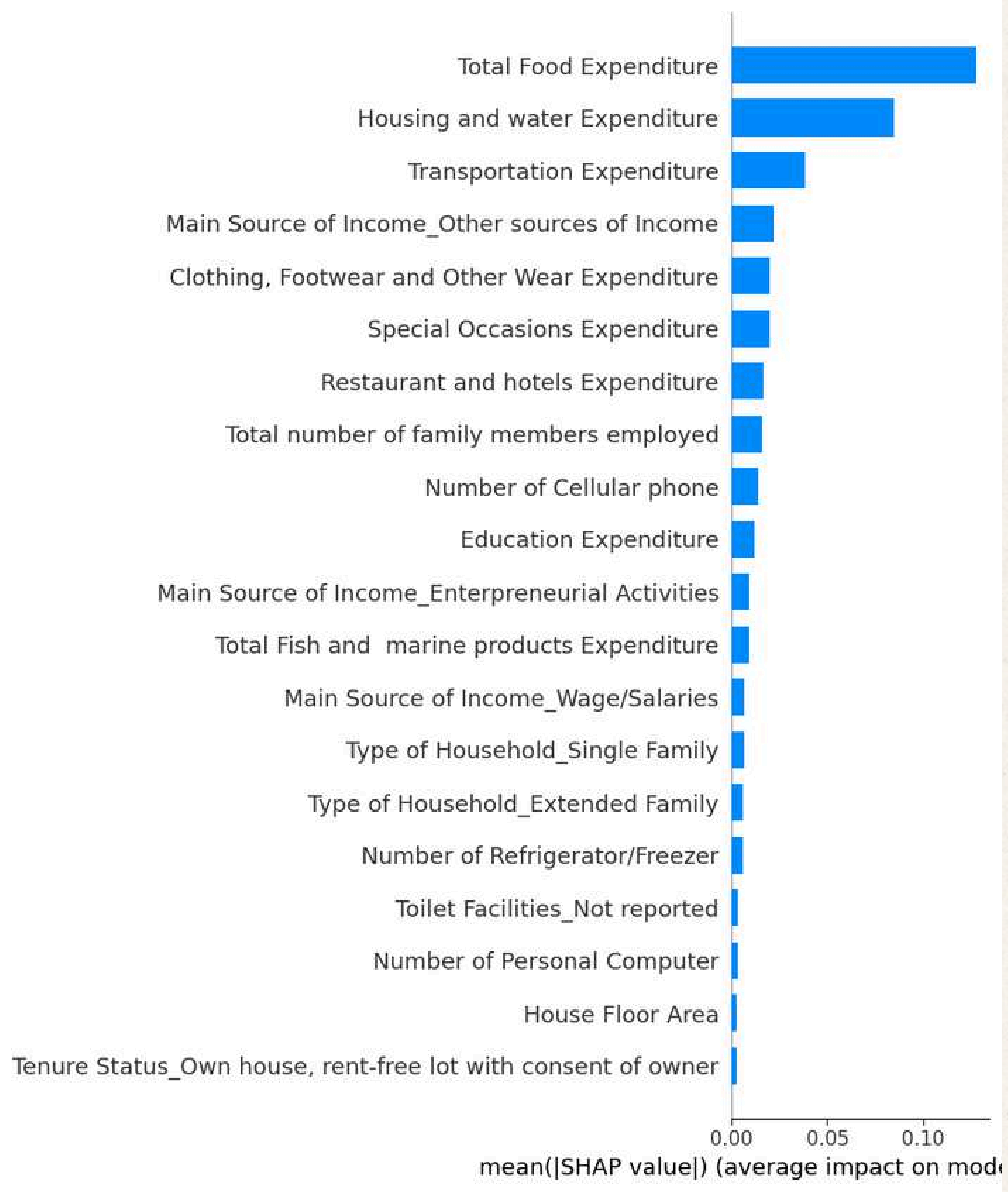
SHAP Analysis

- **Expenditure patterns** are **still** more predictive than income source or demographic.

Total Food and Housing & Water Expenditure remain the undisputed top two features,

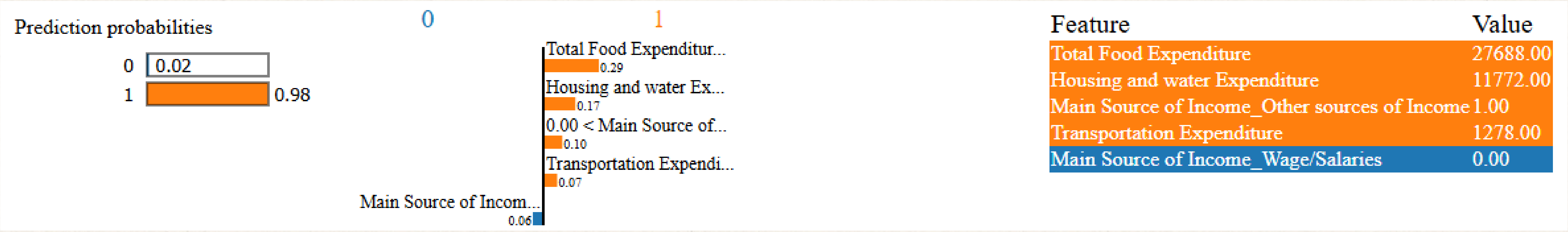
- **Transportation and Income Source** (Other) strengthen their positions

Tuned model better captured mobility and non-traditional income as meaningful class differentiators



Hyperparameter Tuned Model

LIME Analysis



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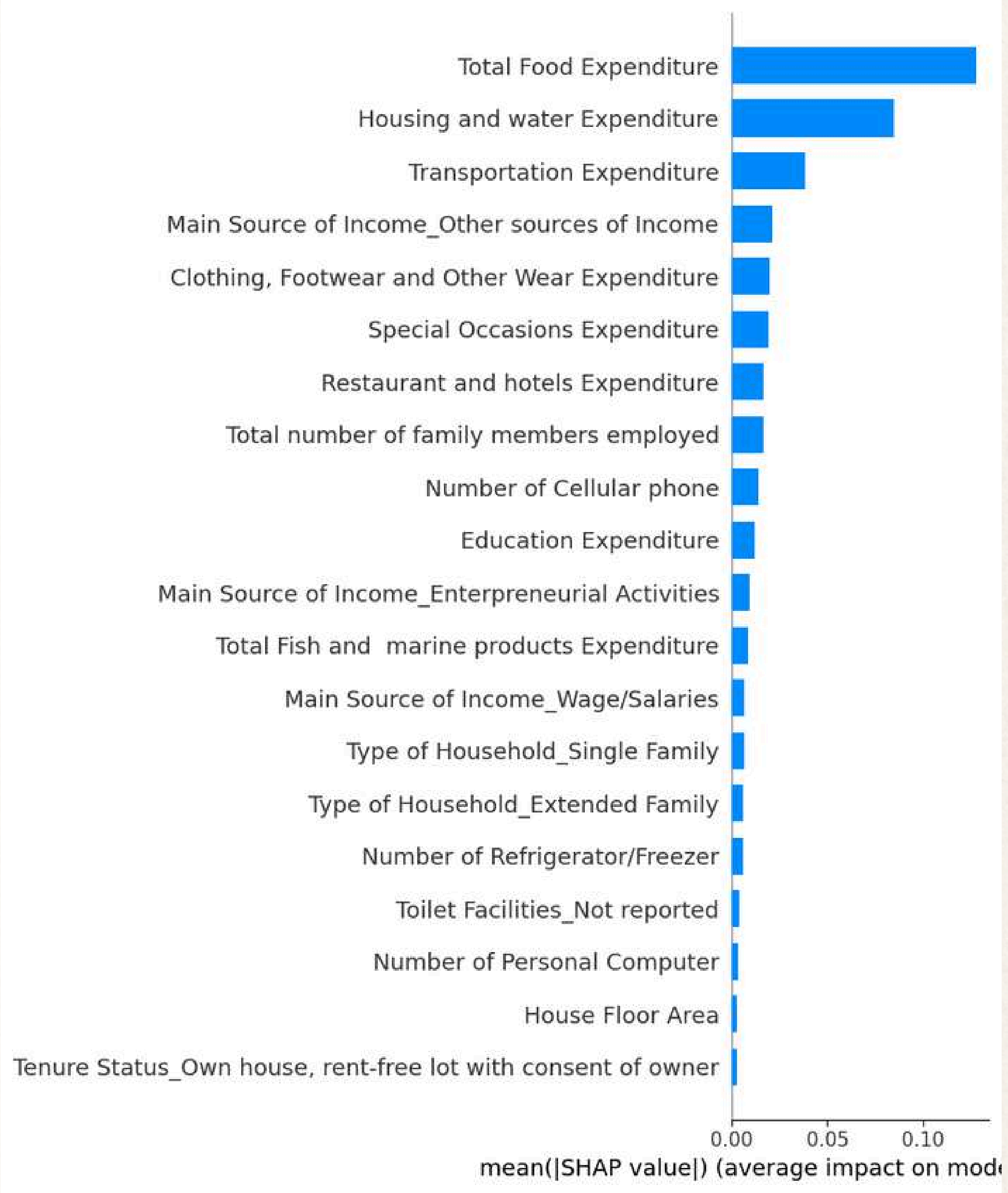
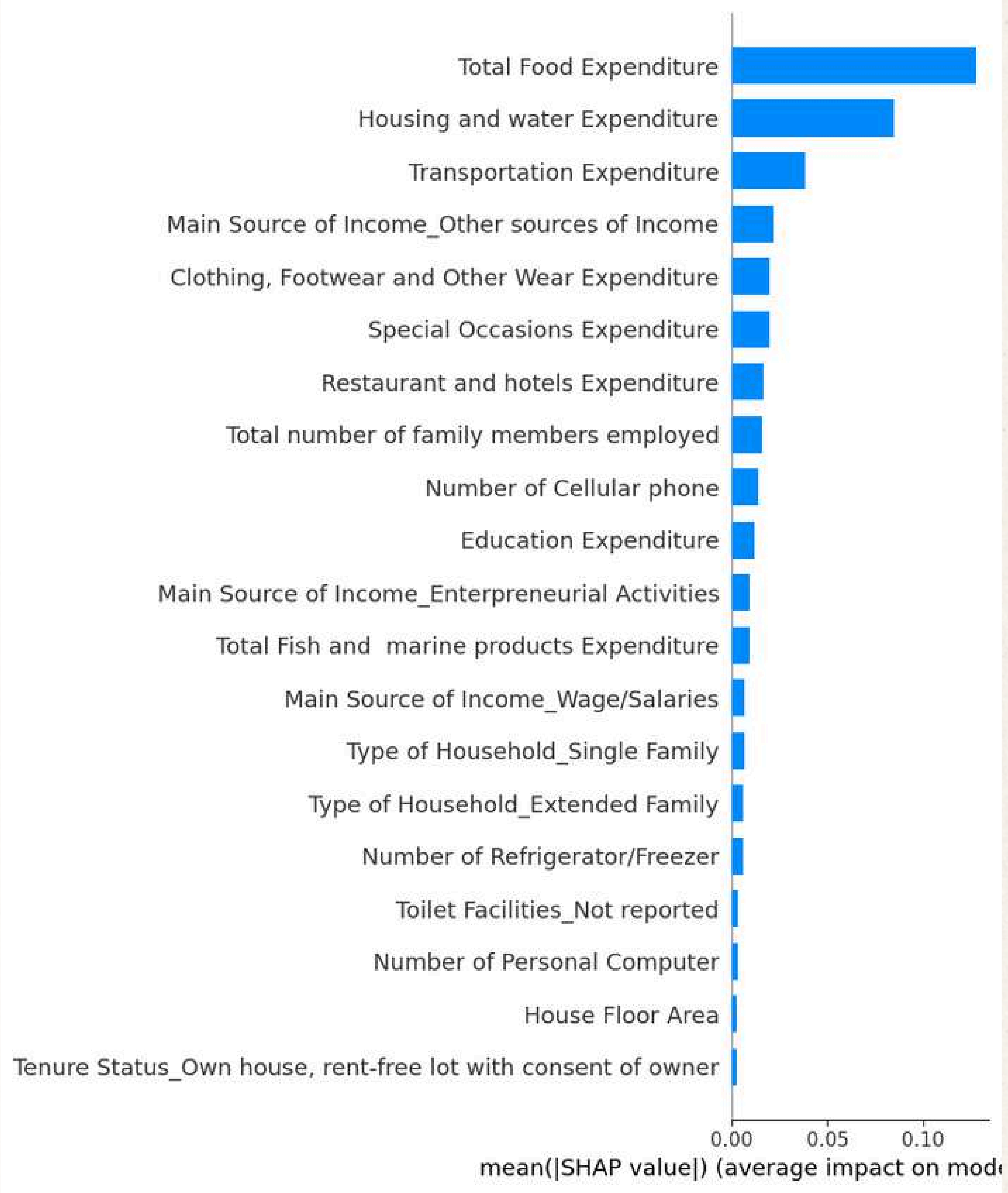
Basic Necessities Matter!

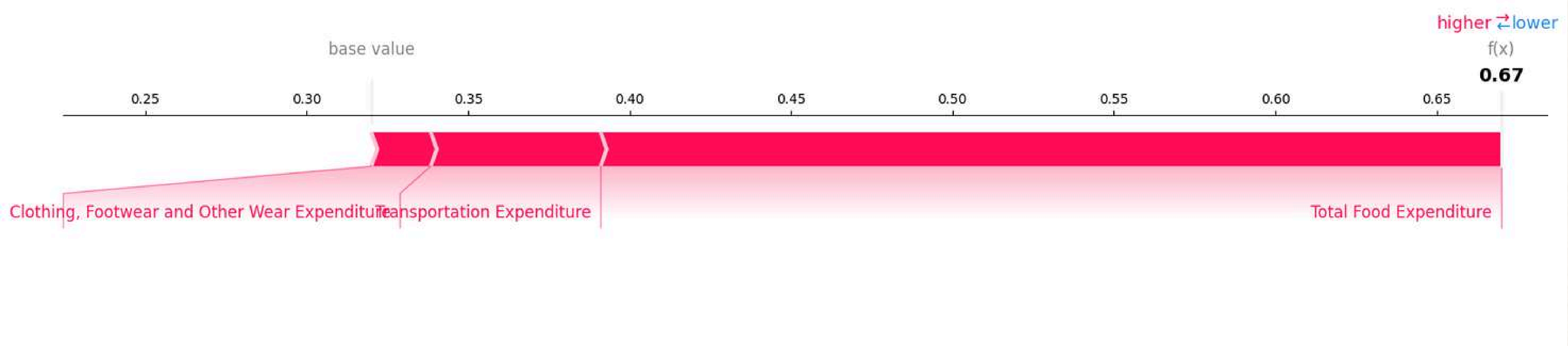




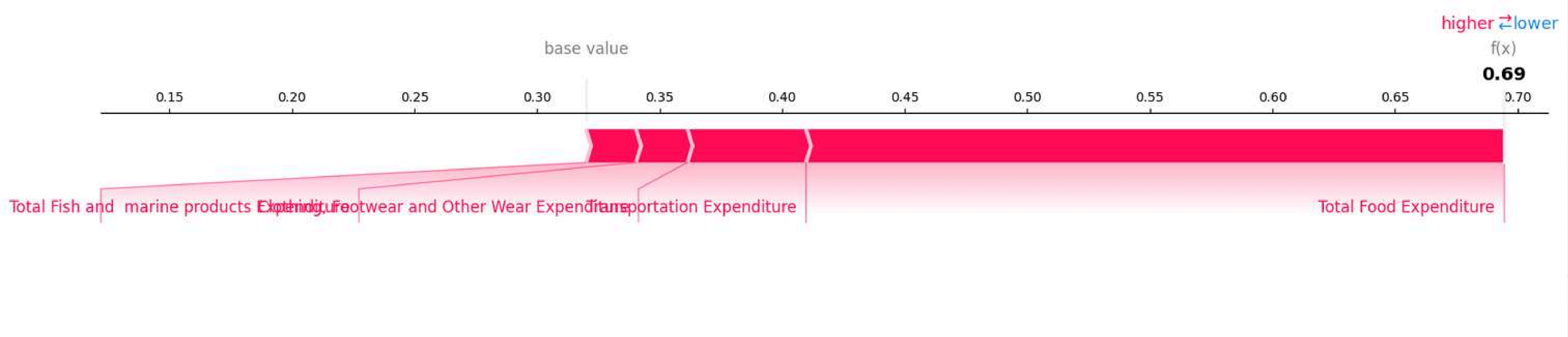
SHAP Baseline

SHAP Hypertuned

Basic Necessities Matter!

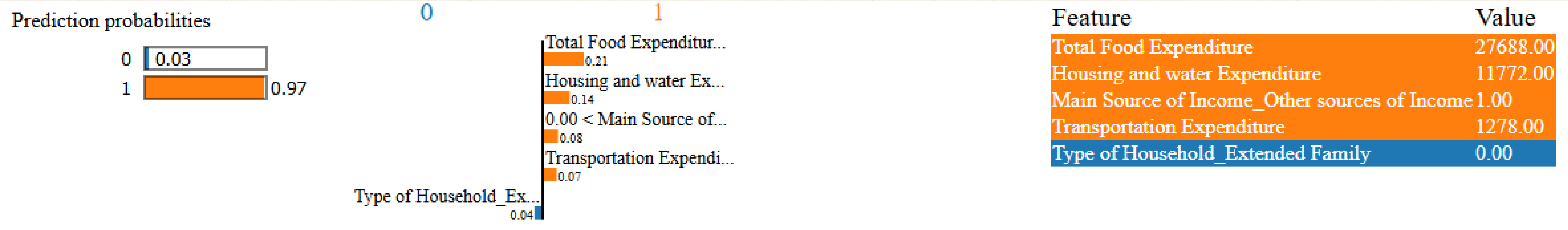


Force Plot Baseline

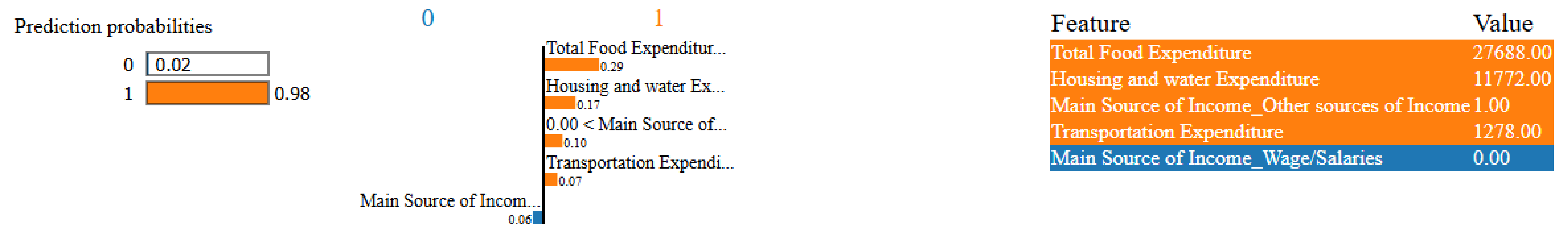


Force Plot Hypertuned

Basic Necessities Matter!

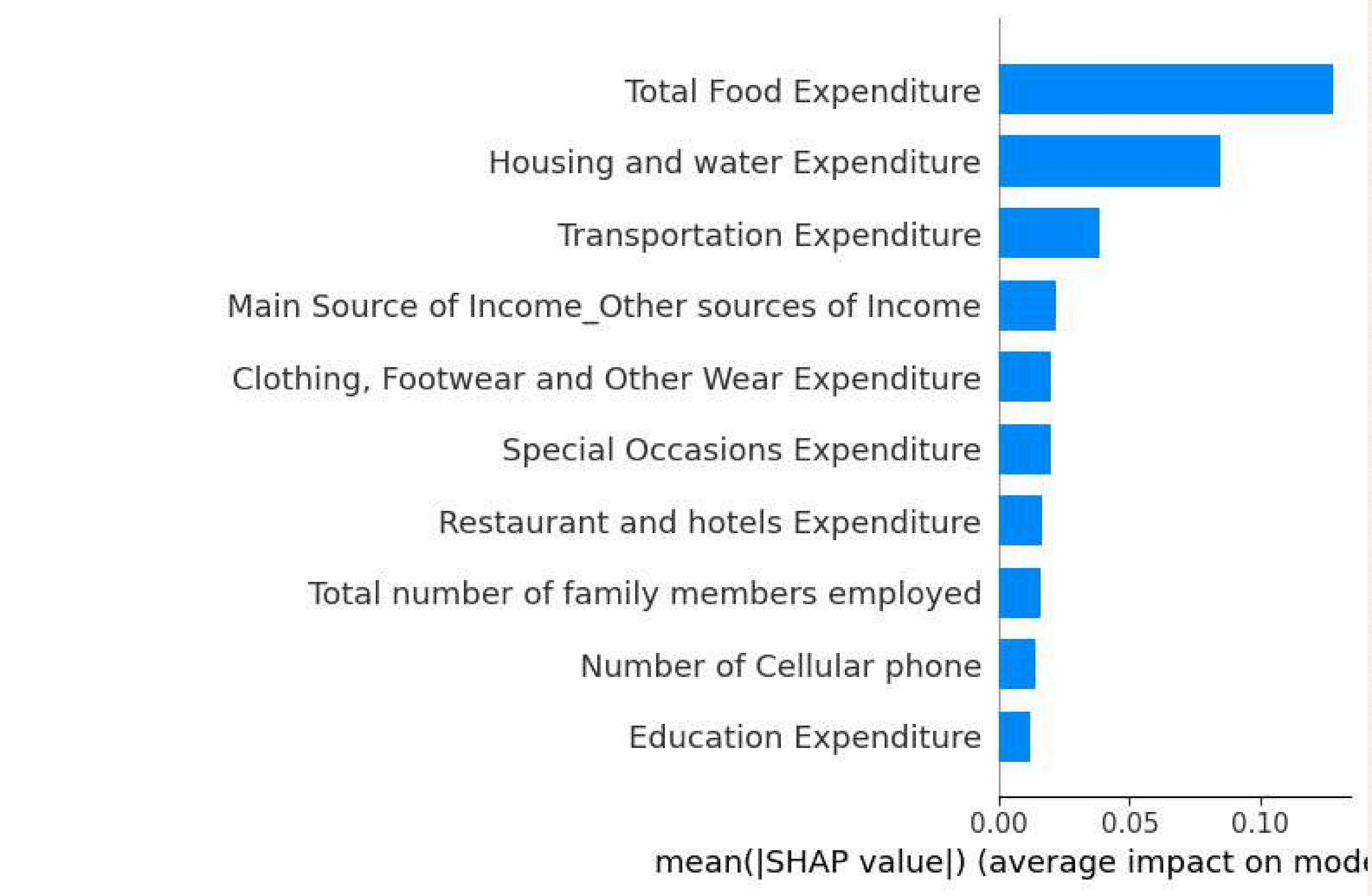


LIME Baseline



LIME Hypertuned

Top 10 Key Factors



#8: Insights and Recommendations

Insights

- **Top Predictors remain the same**, but hyperparameter tuning sharpened the **model's confidence** without drastically changing feature priorities.

- **Expenditure** patterns are the **dominant predictors** of household income class. Specifically, **Total food** and **Housing & Water** Expenditure are the most consistent and powerful predictors across all four analyses

- **Total Food and Housing & Water Expenditure** are the most stable and reliable features

Recommendations

- **Retrain the model using more recent FIES datasets (2018, 2021)** – since expenditure behavior has shifted significantly due to inflation and post-pandemic economic changes
- **Prioritize food, housing, and water expenditure data collection in future surveys or model updates**, as these variables alone carry the most predictive weight and would allow for leaner, more efficient models without sacrificing accuracy
- **Consider removing or consolidating low-importance features** such as income source type, number of televisions, and employment count in future model iterations to reduce dimensionality and improve training efficiency without meaningful loss in accuracy

Q & A